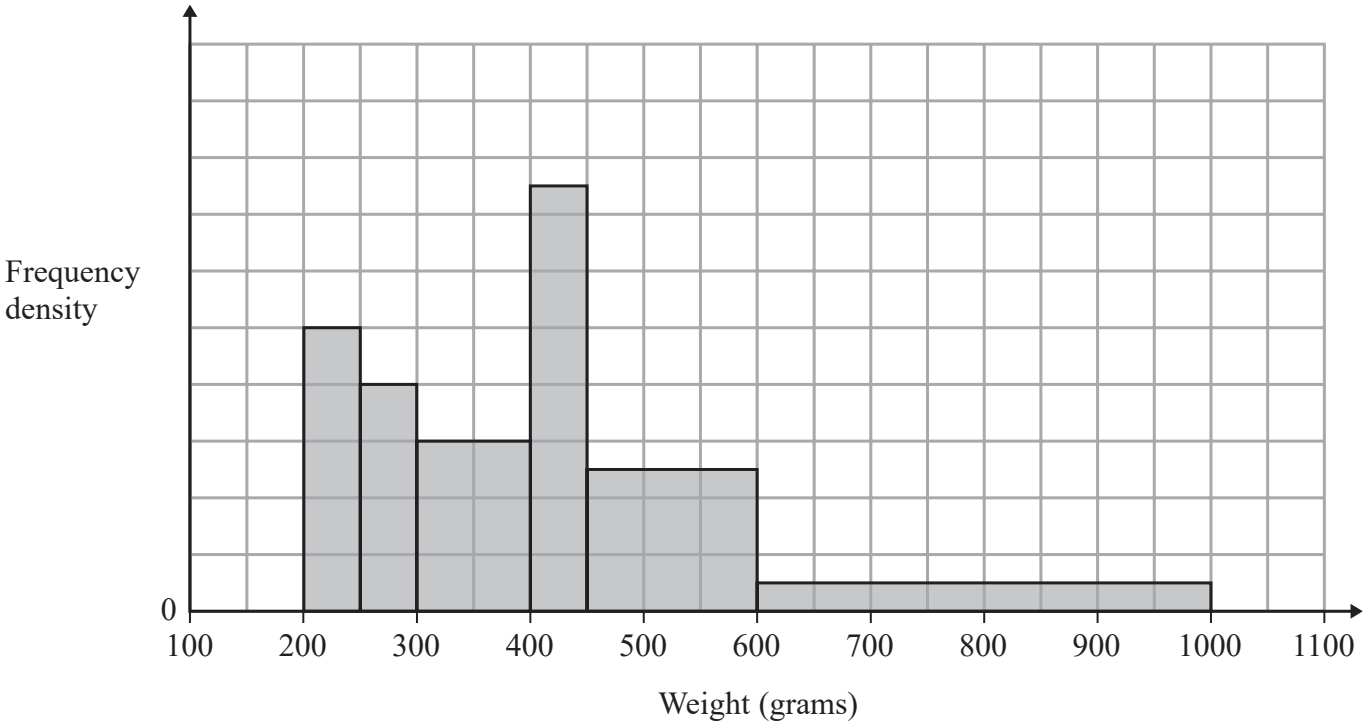
18 The histogram gives information about the weights of some fish.



The number of fish with a weight between 400 g and 450 g is 7 more than the number of fish with a weight between 250 g and 300 g.

(a) Calculate the total number of fish represented by the histogram.

Equal 50g widths: heights differ by 7.5 – 4 = 3.5 squares.

3.5 grid-square areas represent 7 fish: 1 square = 2 fish.

200–250:10; 250–300:8; 300–400:12;

400–450:15; 450–600:15; 600–1000:8.

Total = 10 + 8 + 12 + 15 + 15 + 8 = 68 fish.

1 mark
Calibrate area

1 mark
At least four counts

1 mark
68 fish

.....

(3)

(b) (i) Use the histogram to find an estimate for the median weight.

Half of 68 = 34; cumulative frequency to 400g = 30.

400 + ((34 – 30)/15) × 50 = 413.33... ≈ 413g.

1 mark
Interpolate median

1 mark
About413g

..... g

(2)

(ii) Give a reason why your answer to part (b)(i) is only an estimate.

Individual weights in 400–450g are unknown.

1 mark
Grouped-data estimate

The interpolation assumes an even spread in that class.

(1)

(Total for Question 18 is 6 marks)